

Problem 2: Tracking Targets through the Blind Doppler

This case study is taken from Chapter 9 of [2] and more details can be found here.

Let the true target state at discrete time t_k be defined as

$$\mathbf{x}_k = [x_k \ \dot{x}_k \ y_k \ \dot{y}_k]^T,$$

where x_k and y_k denote the position in Cartesian coordinates and $\dot{x}_k$ and $\dot{y}_k$ are velocities. The polar measurements of the target location at t_k (range r_k and azimuth θ_k) obtained from a radar are conveniently converted into the Cartesian system.

The target motion model is the constant velocity (CV) model, with an appropriate level of process noise to deal with possible maneuvers. In order to simplify analysis we assume a static tracking radar platform. We adopt the following model of target state dynamics:

$$\mathbf{x}_{k+1} = \mathbf{F}_k \mathbf{x}_k + \mathbf{G}_k \mathbf{w}_k$$

where

$$\mathbf{F}_k = \begin{bmatrix} 1 & T_k & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & T_k \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{G}_k = \begin{bmatrix} T_k^2/2 & 0 \\ T_k & 0 \\ 0 & T_k^2/2 \\ 0 & T_k \end{bmatrix}$$

$T_k = t_{k+1} - t_k$, $\mathbf{w}_k$ is a 2×1 white Gaussian process noise vector with covariance matrix

$$\mathbf{Q} = \text{diag}[\sigma_w^2, \sigma_w^2].$$

The measurement equation is given by:

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{w}_k$$

where $\mathbf{z}_k = [X_k \ Y_k \ \dot{r}_k]^T$ is the measurement vector at time t_k . It consists of the Cartesian target position components $X_k = \lambda^{-1} r_k \cos \theta_k$, $Y_k = \lambda^{-1} r_k \sin \theta_k$, where $\lambda = \exp(-\sigma_\theta^2/2)$ and the target range rate measurement $\dot{r}_k$. The error statistics of radar measurements are given in terms of the range standard deviation σ_r , range rate standard deviation $\sigma_{\dot{r}}$, and the azimuth standard deviation σ_θ . With these statistics, the position variances in the respective directions and their cross covariance are as follows:

$$\begin{aligned} \sigma_{X_k}^2 &= (\lambda^{-2} - 2)r_k^2 \cos^2 \theta_k + (r_k^2 + \sigma_r^2)(1 + \lambda^4 \cos 2\theta_k)/2 \\ \sigma_{Y_k}^2 &= (\lambda^{-2} - 2)r_k^2 \sin^2 \theta_k + (r_k^2 + \sigma_r^2)(1 - \lambda^4 \cos 2\theta_k)/2 \\ \sigma_{X_k Y_k}^2 &= (\lambda^{-2} - 2)r_k^2 \cos \theta_k \sin \theta_k + (r_k^2 + \sigma_r^2)\lambda^4 \sin 2\theta_k/2 \end{aligned}$$

The nonlinear measurement function $\mathbf{h}(\mathbf{x}_k)$ is defined as:

$$\mathbf{h}(\mathbf{x}_k) = \begin{bmatrix} x_k \\ y_k \\ \frac{x_k \dot{x}_k + y_k \dot{y}_k}{\sqrt{x_k^2 + y_k^2}} \end{bmatrix}$$

Measurement noise $\mathbf{w}_k$ is a 3×1 zero-mean Gaussian noise vector with covariance matrix

$$\mathbf{R}_k = \begin{bmatrix} \sigma_{X_k}^2 & \sigma_{X_k Y_k}^2 & 0 \\ \sigma_{X_k Y_k}^2 & \sigma_{Y_k}^2 & 0 \\ 0 & 0 & \sigma_{\dot{r}}^2 \end{bmatrix}$$

The assumption is that $\mathbf{v}_k$ and $\mathbf{w}_k$ are independent. A simple model of a 2D radar is assumed for the purpose of studying the problem of tracking targets occasionally hidden in the blind Doppler. The probability of detection according to this model is:

$$P_D(\mathbf{x}_k) = \begin{cases} 1, & \text{if } \left| \frac{x_k \dot{x}_k + y_k \dot{y}_k}{\sqrt{x_k^2 + y_k^2}} \right| \geq L_0 \\ 0, & \text{otherwise} \end{cases}$$

where L_0 is the limit of the blind doppler. For this problem assume $L_0 = 100 \text{ km/h}$. You can consider the target starting from 60Km north of the radar system and moving towards it with a speed 800 km/h for 90 seconds, followed by a turn to its right for 25 seconds and finally a turn to its left to continue moving towards the radar. Moreover, $\sigma_w = 9.81/\sqrt{T_k}$. Assume $T = 2s$, $\sigma_r = 250 \text{ m}$, $\dot{\sigma}_r = 3 \text{ m/s}$ and $\sigma_\theta = 1^\circ$.